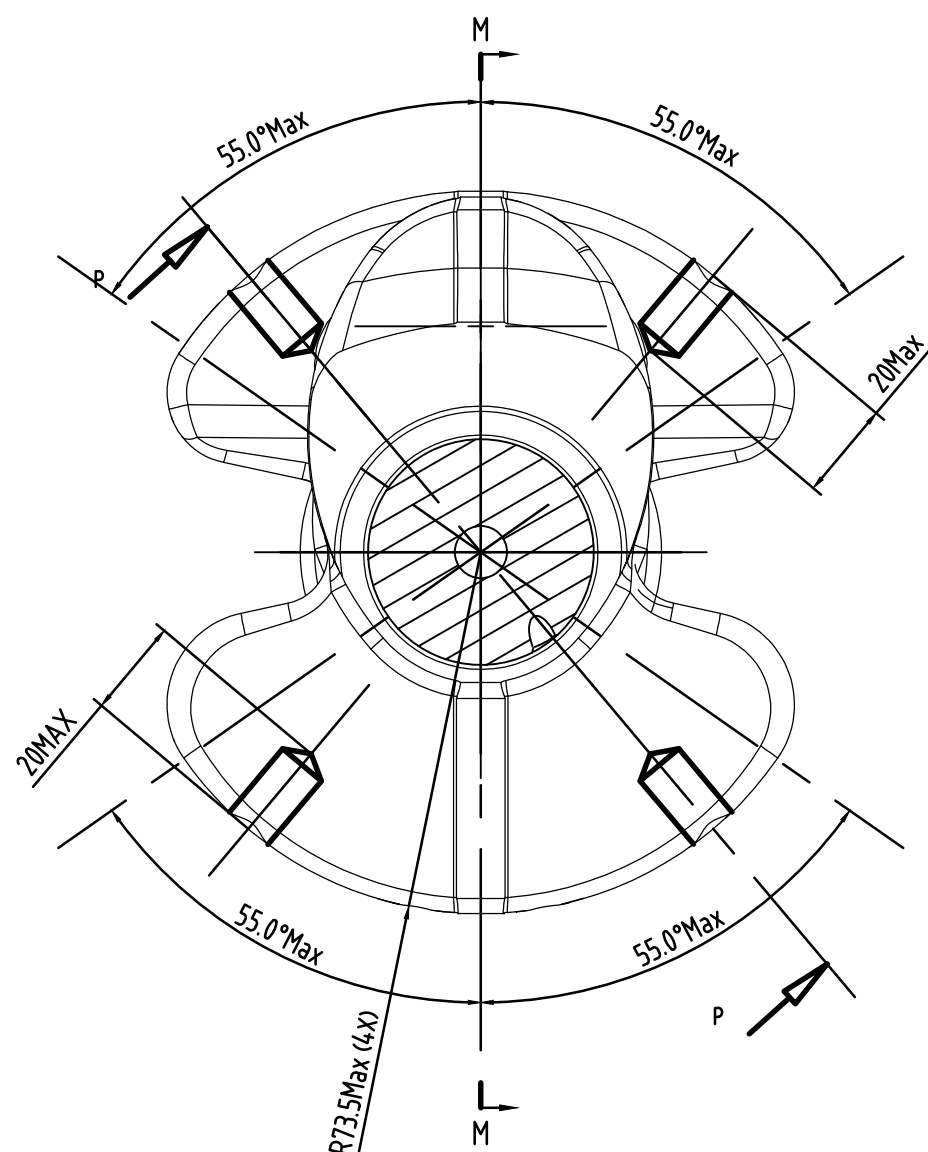
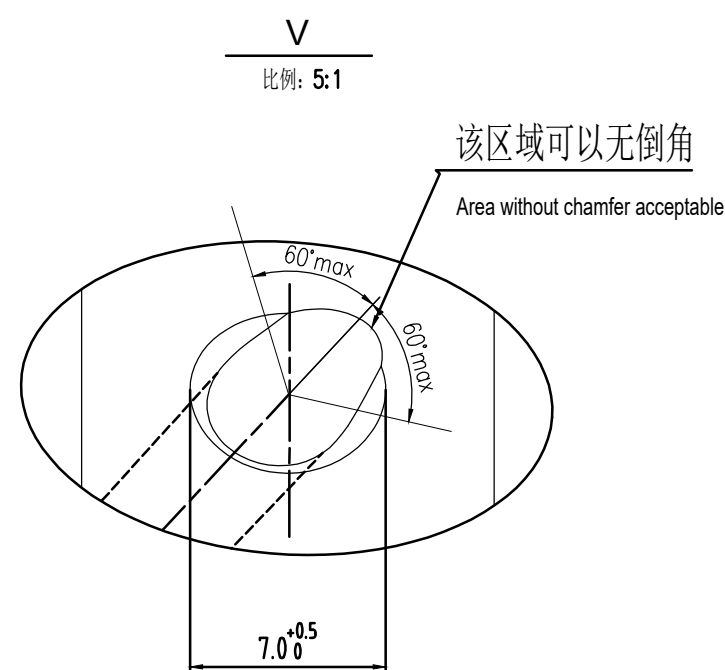
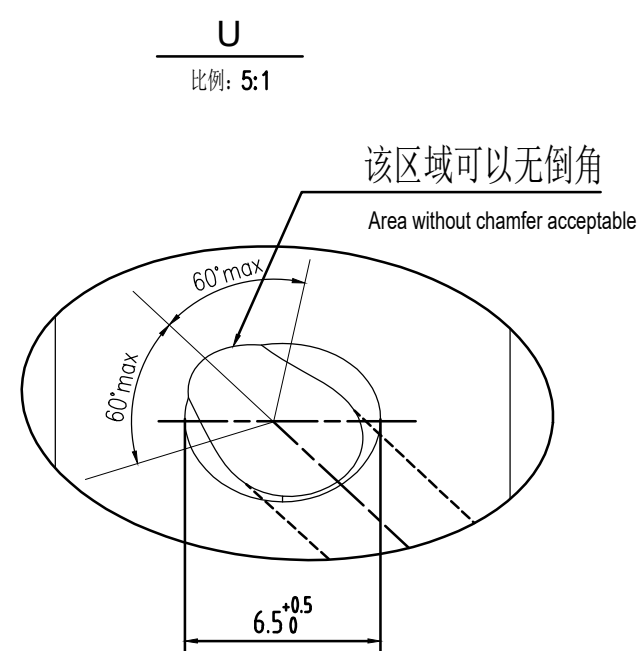
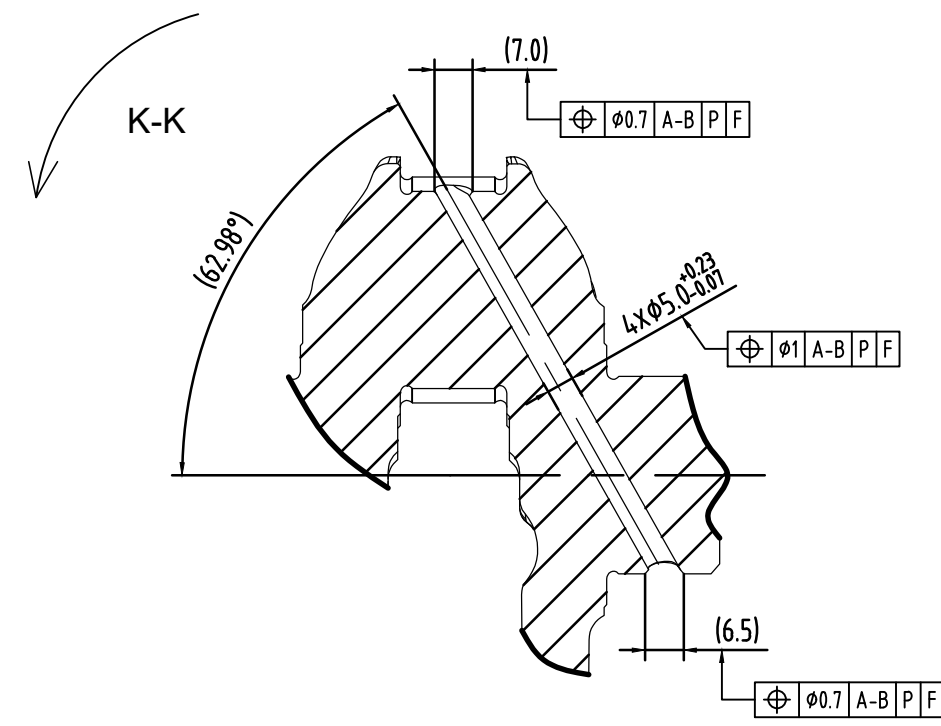


1. The 1,4,5,8 CWs are allow to be drilled balancing holes. The number of holes are less than 3 in each CW, the area will under 55 degree, not drill out and the depth is 20 max.
2. There are not balancing hole in other CWs.



滚压头负载
CLAMPING

F

F_x

X°

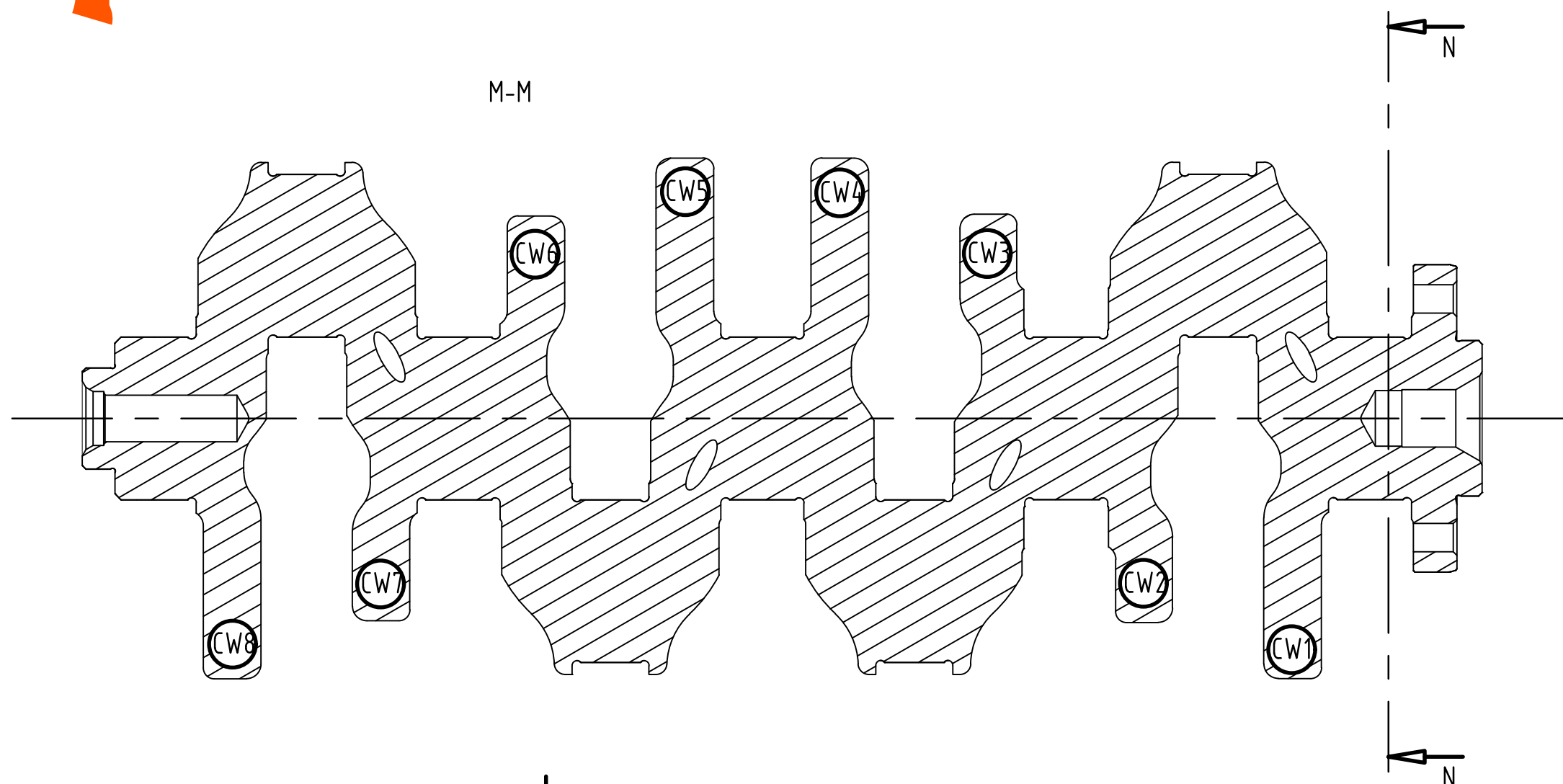
ROLLING FORCE

主轴承 & 压杆
MAIN S & PINS

ROLLING FORCE

滚压头负载

A	B	C	D	E	G	H	I	K	M	N	P	Q	R	S	T	U
44,974	44,975	44,976	44,977	44,978	44,979	44,980	44,981	44,982	44,983	44,984	44,985	44,986	44,987	44,988	44,989	44,990

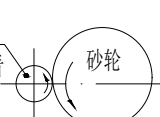


- 加工表面不允许有毛刺、尖角硬凹陷、气孔、涌浪、夹渣及非金属材料夹杂等缺陷。
- 曲轴应进行动平衡试验,不平衡量 $\leq 20\text{g}\cdot\text{cm}$,考虑定位销的不平衡补偿值 $\pm 6\pm 5.9\sqrt{R33\text{mm}}=19.5\text{g}\cdot\text{cm}$ 。
- 未注公差的角度和尺寸公差的公差按GB/T1804—m,未注形位公差按GB/T1184—H。

② ① 曲轴精加工后应仔细剔除油泥和各位的金属屑及杂物,曲轴整体清洁度 $\leq 7\text{mg}$,每毫米处中直径为 $150\sim 600\mu\text{m}$ 的金属颗粒数不能超过100粒(600 μm >100P/mg Max.>150 um)无大于 $600\mu\text{m}$ 直径的金属颗粒,清洁度检测按GB16232标准进行。

5. 曲轴按从小头前端(正时端)观察的逆时针旋转方向,进行对主轴承、连杆轴颈的磨削,抛光应与磨削方向相反。

曲轴磨削时



曲轴抛光时



发动机旋转方向



- 曲轴按GB6729—2007《内燃机曲轴、凸轮轴磁粉探伤》进行磁力探伤,磁力探伤后应作退磁处理。
- 其余按GB16232—2005《发动机曲轴技术条件》的规定执行。
- 材料38MnSiV5,成分控制: C:0.36~0.40%;S:0.50~0.80%;Mn:1.20~1.50%;P:0.02%, S:0.045~0.065%;Cr:0.10~0.20%;Mo≤0.06%;Ni≤0.15%;Al:0.01~0.04%;V:0.15~0.140%, Ti≤0.02,Cu≤0.30,N≥0.0125%, 散件理化性能要求: 1005MPa>抗拉强度≥845MPa,断后伸长率≥12%,屈服强度>580MPa,硬度255~302HB,金相组织: 珠光体60%~90%,铁素体40%~10%,晶粒度3~6级;

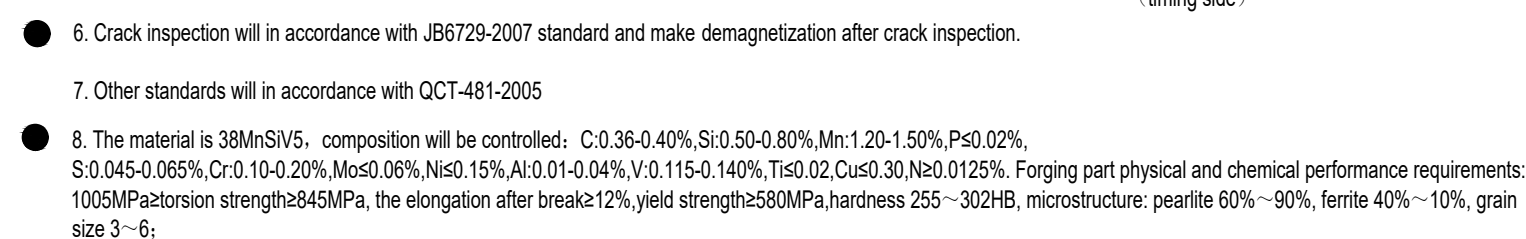
③ ① 1. There is not burr, sharp corner, impurity, pore and so on in machining surface.

② 2. The unbalancing $\leq 20\text{g}\cdot\text{cm}$, consider the unbalancing compensation value in pin hole is $\Phi 8[5.9\sqrt{R33\text{mm}}=19.5\text{g}\cdot\text{cm}]$.

3. The tolerance of undefined linear and angle will in accordance with GB/T1804-m, the undefined geometric tolerance will in accordance with GB/T1184-H.

② ① 4. The cleanliness of whole crankshaft is $\leq 7\text{mg}$, metal particle(diameter is $150\sim 600\mu\text{m}$) counts of each mg less than 100 (600 μm >100P/mg Max.>150 um), metal particle counts 0 P Max.>600 um. According to ISO 16232.

5. From front side (timing side), the grinding direction is same as clock direction, and the polishing direction is opposite as grinding direction.



1、测量抽取3个截面（截面1、截面2、截面3），测量时，曲线旋转半圆（180°），每个截面测量30个点，取平均值分别为 θa 、 θb 、 θc

$$\theta a = (\theta a1 + \theta a2 + \dots + \theta a30) / 30$$

$$\theta b = (\theta b1 + \theta b2 + \dots + \theta b30) / 30$$

$$\theta c = (\theta c1 + \theta c2 + \dots + \theta c30) / 30$$

2、抽取的分组测量值为 $(\theta a + \theta b + \theta c) / 3$

1. Measure three sections (section 1, 2, 3) during measurement, rotate crankshaft 180°, measure 30 points in each sections, then calculate the average of $\bar{O}a, \bar{O}b, \bar{O}c$.
2. the measurement result of grouping is $(\bar{O}a + \bar{O}b + \bar{O}c) / 3$

2. the measurement result of grouping is $(Oa+Ob+Oc)/3$